

JATIN

SNOWFLAKE ENGINEER , 7+

Summary: Experienced Snowflake Data Engineer and AI Specialist with 7+ years in data engineering and cloud data platforms. Proven track record delivering enterprise-scale Snowflake solutions — including performance optimization, complex query tuning, and advanced AI-powered applications using Snowflake Cortex AI and Cortex Agents. Deep expertise in Medallion Lakehouse Architecture, ELT pipeline engineering, and LLM-based data solutions. Adept at optimizing multi-hundred-million-row datasets for sub-second analytics and enabling self-serve AI capabilities for business teams.

SKILLS

Snowflake: Performance Optimization, Query Tuning, Architecture, Clustering Keys, Partitioning, Cost Optimization
Snowflake AI: Cortex AI, Cortex Agents, LLM-Powered Solutions, AI/ML Application Development & Testing
Data Engineering ELT/ETL Pipelines, Medallion Architecture (Bronze/Silver/Gold), Delta Lake, Incremental & CDC Loads
Cloud Platforms Azure Databricks, Azure Data Factory, ADLS Gen2, GCP BigQuery, AWS (S3, Glue), Snowflake
Languages & Tools: Python, PySpark, SQL (SparkSQL, BigQuery SQL, HiveQL), DuckDB, Pandas, Shell
Orchestration: Apache Airflow, Databricks Workflows, Azure DevOps, CI/CD Pipelines, Terraform (IaC), Git

PROJECTS

Project: Procurement Portal | Snowflake + BigQuery ELT & AI Analytics

- Snowflake · GCP BigQuery · ELT · Python · Azure Databricks · Cortex AI · SQL Performance Optimization
- Led end-to-end ELT migration and ingestion pipelines into Snowflake and GCP BigQuery for an enterprise Procurement Data platform powering the Opportunity Scanner module used by commercial ops teams.
- Designed scalable ELT workflows with schema consistency, data quality validation, and full lineage traceability across the Gold layer — ensuring reliable data for AI-driven procurement analysis.
- Optimized Snowflake and BigQuery query performance through partitioning strategies, clustering keys, and query plan tuning — achieved sub-3-second response on 500M+ row procurement datasets for real-time analytics.
- Applied Snowflake Cortex AI capabilities to power intelligent anomaly detection and spend categorization within the Opportunity Scanner — surfacing outliers in vendor risk scores and contract compliance automatically.
- Built data quality validation layers with automated anomaly detection for procurement KPIs — flagging spend outliers prior to Gold layer promotion, reducing manual QA effort significantly.

Project: Pharma Commercial Analytics Platform | Snowflake Lakehouse + Cortex AI

- Snowflake · Cortex AI · LLM Solutions · Delta Lake · Azure Databricks · PySpark · Medallion Architecture · CI/CD

Ingestion: Auto Loader, Apache NiFi, Spark Streaming, REST APIs, SFTP, JDBC, Delta Live Tables
Databases: Snowflake, BigQuery, MySQL, PostgreSQL, MongoDB, SQLite3, Hive
Domains: Pharma Commercial Analytics, Procurement, Healthcare (FHIR/HL7), Accounting/ERP
Soft Skills: Architecture Workshops, Cross-functional Leadership, Technical Mentorship, Agile/Scrum

- Architected an end-to-end Lakehouse platform ingesting sale, prescription (Rx), and HCP engagement data from 15+ upstream sources — with Snowflake as the serving and AI layer for commercial analytics.
- Designed and implemented Medallion Architecture (Bronze → Silver → Gold): Bronze for raw multi-format ingestion (JSON, XML, CSV, HL7), Silver for cleansed datasets, Gold for KPI-ready aggregations powering executive dashboards.
- Integrated Snowflake Cortex AI to deliver LLM-powered natural language querying on Gold-layer certified tables — enabling business users and commercial ops teams to query data via plain English, reducing ad-hoc SQL requests to the data team by 60%.
- Built real-time and batch PySpark pipelines for processing high-volume commercial Rx data feeds, reducing pipeline latency from 6 hours to under 30 minutes with ACID transactions and Z-ordering.
- Designed self-serve analytics layer using Snowflake SQL Warehouses and PowerBI — enabling business users and data scientists to query certified Gold tables without engineering dependency.
- Implemented CI/CD pipelines using Azure DevOps with automated unit tests, notebook version control via Git, and multi-environment (dev/staging/prod) workspace promotion workflows.
- Provisioned and managed infrastructure via Terraform (IaC) — clusters, job configurations, secret scopes, and ADLS Gen2 mount points — enabling repeatable environment setup and compliance.
- Mentored a team of 4 junior data engineers — conducted weekly code reviews, architecture design sessions, and hands-on POC workshops on Delta Live Tables and Auto Loader features.

Project: Cloudera → Cloud Migration | Medallion Architecture & Snowflake Optimization

- Snowflake · Azure Databricks · Delta Lake · HiveQL → SparkSQL · Airflow · Terraform · Query Optimization
- Led end-to-end migration from Cloudera (Hive/HDFS) to cloud-native Snowflake and Azure Databricks — translated 200+ HiveQL queries into optimized SparkSQL/PySpark transformations with partitioning, caching, and broadcast join strategies.
- Implemented full Medallion Architecture (Bronze raw → Silver cleansed → Gold business-ready) with incremental load (Change Data Capture) and full snapshot patterns, ensuring zero data loss and full auditability across Snowflake and Delta Lake layers.

- Tuned Snowflake query performance extensively — analyzed execution plans, optimized joins and aggregations, applied clustering keys and result caching — achieving significant reduction in query runtime on large-scale datasets.
- Built Python-based Airflow DAGs for end-to-end orchestration with retry logic, SLA alerting, cross-DAG dependencies, and dynamic task mapping across 20+ pipeline stages.
- Achieved 55% reduction in compute costs through auto-scaling policies, spot instance strategies, and query optimization using Delta's liquid clustering and Snowflake warehouse sizing strategies.
- Configured Private Endpoints, VNet integration, and secret management via Azure Key Vault to ensure zero-trust data access and enterprise security compliance.
- Provisioned all infrastructure via Terraform — workspaces, cluster policies, instance pools, job clusters — enabling version-controlled, reproducible environment deployments across dev/staging/prod.

Project: Delta Live Tables | Real-Time Streaming Data Platform with Cortex AI POC

- Azure Databricks · Delta Live Tables · Spark Structured Streaming · Unity Catalog · Azure Event Hubs · Snowflake Cortex AI POC
- Architected a real-time streaming data platform using Databricks Delta Live Tables (DLT) to ingest, validate, and transform high-velocity event streams — processing 10M+ events/day across multiple business domains.
- Designed DLT declarative pipelines with expectations-based data quality enforcement (quarantine pattern) — automatically routing bad records to error tables while ensuring clean data flows into Silver and Gold layers without pipeline interruption.
- Conducted POC on Snowflake Cortex AI and Databricks AI/BI Genie for self-serve natural language analytics — enabling business users to query certified Delta tables via LLM-powered interfaces, reducing ad-hoc SQL requests by 60%.
- Implemented Auto Loader with schema evolution for file-arrival-based ingestion from ADLS Gen2, handling schema drift across JSON, Avro, and Parquet formats without manual intervention.
- Leveraged Unity Catalog for fine-grained access control, data lineage tracking, and centralized governance — enabling column-level security and audit logging for compliance.

- Tuned Spark Structured Streaming performance via trigger intervals, watermarking, stateful aggregations, and RocksDB state store for low-latency, exactly-once processing guarantees.

Project: Databricks Lakehouse POC | Unity Catalog + MLflow + Snowflake Feature Store

- Azure Databricks · Unity Catalog · MLflow · Feature Store · Snowflake · Python · CI/CD · Azure DevOps
- Led a Lakehouse POC evaluating Unity Catalog, MLflow Model Registry, and Feature Store as a unified platform for analytics and ML — presented findings and adoption roadmap to client's architecture board.
- Designed a Feature Store pipeline to compute and serve 80+ reusable ML features (HCP segmentation, territory performance, Rx trend signals) from Gold-layer Delta tables — enabling consistent features across training and inference.
- Set up MLflow end-to-end model lifecycle management — experiment tracking, model versioning, staging/production promotion workflows, and REST API serving via Model Serving endpoints.
- Built CI/CD pipeline using Azure DevOps + Databricks Asset Bundles (DABs) — automated testing of notebooks, DLT pipelines, and MLflow models with environment-specific parameter injection and one-click deployment to production.
- Performed performance benchmarking — compared Photon engine vs standard runtime on TPC-DS workloads, demonstrated 3x query speedup on Gold-layer aggregations justifying Photon cluster adoption for BI workloads.
- Delivered technical mentorship sessions on Delta Lake internals (file compaction, VACUUM, OPTIMIZE, Z-order), Unity Catalog migration, and DLT best practices to onshore and offshore teams.

Project: Healthcare Clinical Data Platform | FHIR/HL7 ETL & Snowflake

- Healthcare · FHIR/HL7 · Azure Databricks · Snowflake · Apache NiFi · MongoDB · PySpark · MySQL · HIPAA
- Designed and deployed a centralized clinical data ingestion platform aggregating patient data from 10+ hospital and payer sources — reduced processing time by 45% and improved data accuracy by 20%.
- Implemented HL7/FHIR-compliant data transformation mapping raw MongoDB clinical records to FHIR R4 resources (Patient, Observation, Encounter, Diagnostic Report, Medication Request) ensuring full interoperability and HIPAA compliance.

- Architected a clinical datamart on Snowflake with optimized schema design (star schema with SCD Type 2) for longitudinal patient history — supporting population health analytics and care coordination reporting.
- Built Apache NiFi flows for real-time clinical data ingestion with routing logic, schema validation, error queuing, and dead-letter handling across multi-source hospital feeds — handling 2M+ daily clinical events.
- Engineered an automated audit and reconciliation pipeline to detect and recover failed CPF (Care Plan Flow) records from MongoDB — saving 20+ hours/week of manual intervention and reducing data entry errors by 80%.
- Designed PHI masking, row-level security, and audit trail logging across the pipeline aligned with HIPAA technical safeguards and organizational data governance policies.

Project: Accounting ERP Migration | QuickBooks & Sage ELT Pipelines

- QuickBooks · Sage 100/300 · Python · DuckDB · Azure Databricks · REST APIs · SQL Server
- Designed full-scope ELT pipelines to migrate accounting data from QuickBooks Desktop, Sage 100, and Sage 300 to QuickBooks Online — covering Chart of Accounts, Invoices, Vendors, Journal Entries, and multi-year Transactions via Python and REST APIs.
- Leveraged Azure Databricks for batch processing of 5M+ historical transaction records with parallel transformation jobs, ensuring scalability and reliability across multi-year financial data migrations.
- Used DuckDB for high-speed in-process validation — cross-system trial balance comparisons, debit/credit integrity checks, and pre-load financial reconciliation without shipping data to a remote warehouse.
- Produced automated audit trails and reconciliation dashboards with before/after financial snapshots and transformation logs for CFO sign-off — reduced migration sign-off cycle from 2 weeks to 3 days.

